Chapter 4 Circulation and Excretion

Lesson 4-1

Heart and Blood

A healthy heart keeps your body strong!





Try

1 (1) What is the organ that works like a pump to send blood to the whole body?

The heart.

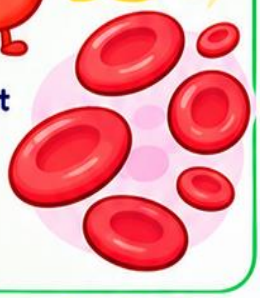
2 (2) What are the important things that blood carries to the whole body? Choose all correct answers from below.

- A: Oxygen
- B: Carbon dioxide
- C: Nutrients
- D: Muscles



3 (3) Why does your pulse become faster after exercising? Choose the correct answer from below.

- A: To let the blood rest slowly.
- B: To carry more oxygen and nutrients to the whole body quickly.
- C: Because there are not enough nutrients.



Exercise

1 (1) Which statement correctly explains the heart? Choose the correct answer from below.

- A: The heart plays a role in digesting the food we eat.
- B: The heart is constantly beating to pump out blood.
- C: The heart plays a role in storing nutrients.



2 (2) What do we call the movement of the heart pumping out blood?

A heartbeat.

3 (3) Fill in the blanks with the correct words.

The blood pumped out from the heart carries (① oxygen) and nutrients to the whole body, collects (② carbon dioxide) and waste produced in the body, and returns to the heart again. The blood that has returned to the heart is sent to the (③ lungs) to exchange (① oxygen) and (② carbon dioxide).









Chapter 4

Lesson 4-1

Heart and Blood

A healthy heart keeps the blood flowing to all parts of your body!



A. Choose the Correct Answer

Choose the best answer.

1 Which organ works like a pump to send blood around the whole body?



a) Lungs
b) Heart
c) Stomach
d) Small intestine

2 Which substances are carried by the blood to the body cells?



a) Oxygen and nutrients
b) Carbon dioxide and waste
c) Saliva and water
d) Carbon dioxide and nutrients

3 Where is blood sent to release carbon dioxide and take in fresh oxygen?



a) Stomach
b) Liver
c) Lungs
d) Small intestine

4 Which part of the circulatory system carries blood from one part of the body to another?



a) Blood vessels
b) Alveoli
c) Bronchi
d) Intestines

5 After delivering oxygen and nutrients to the body cells, blood collects:



a) oxygen and nutrients.
b) carbon dioxide and wastes.
c) food and water.
d) saliva and oxygen.

B. Write True or False

Write True (T) or False (F).

1  The heart is a strong muscular organ. **(T)**

2  Blood carries nutrients absorbed from the small intestine to body cells. **(T)**

3  Blood stays in the heart and does not travel around the body. **(F)**

4  The circulatory system includes the heart, blood vessels, and blood. **(T)**

5  Oxygen-rich blood is pumped from the heart to different parts of the body. **(T)**

C. Complete the Following

Fill in the blanks with the correct words.



1. The heart works like a **pump** that keeps blood moving around the body.
2. Blood carries oxygen and **nutrients** to the body cells.
3. Carbon dioxide produced by body cells is collected by the **blood**.
4. Blood flows around the body through tubes called blood **vessels**.
5. Blood returning from the body is sent to the **lungs** to exchange carbon dioxide for oxygen.

Healthy Hearts · Brighter Lives!



Chapter 4

Lesson 4-1

Heart and Blood

A healthy heart keeps the blood flowing to all parts of your body!



D. What Happens When...?

Write what happens.

1. You exercise and your body needs more oxygen?
   Your heart beats faster and pumps more blood around your body.
2. Oxygen-rich blood leaves the heart and travels to the body cells?
   It delivers oxygen and nutrients to the body cells.
3. Blood reaches the body cells after carrying oxygen and nutrients?
   The cells use the oxygen and nutrients for energy and growth, and the blood picks up carbon dioxide and wastes from the cells.
4. Blood containing carbon dioxide returns to the heart?
   It brings carbon dioxide (waste gas) back to the heart.
5. Blood reaches the lungs?
   It releases carbon dioxide and picks up fresh oxygen.



E. Give Reasons for the Following

Write the reason.

1. The heart beats continuously throughout a person's life.
   Because it pumps blood all the time to keep the body alive.
2. The heart beats faster during exercise.
   Because the body needs more oxygen and nutrients, so the heart pumps more blood to deliver them quickly to all parts of the body.
3. Blood must circulate to all parts of the body.
   Because it carries oxygen and nutrients to the body cells and removes wastes.
4. Blood is sent from the heart to the lungs.
   To release carbon dioxide and pick up fresh oxygen.
5. Blood returns to the heart after passing through the body.
   Because it has delivered oxygen and nutrients and picked up carbon dioxide (waste).



Chapter 4 Circulation and Excretion

Lesson 4-2

How do Kidneys Work?

Healthy kidneys keep your body clean and healthy!





A. Choose the Correct Answer

Choose the best answer.

1 Which organs are mainly responsible for removing wastes from the blood?

a) Lungs
b) Kidneys
c) Stomach
d) Heart

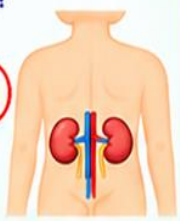


2 What is the main function of the kidneys?

a) To pump blood around the body
b) To filter the blood and remove wastes
c) To store urine
d) To carry urine out of the body

3 Where are the kidneys located in the human body?

a) In the chest
b) In the lower part of the back
c) Inside the head
d) In the neck



4 Which organ stores urine until it is ready to leave the body?

a) Kidney
b) Ureter
c) Bladder
d) Urethra



5 A person's kidneys are working properly. Which result would you expect?

a) Wastes remain in the blood.
b) Cleaned blood returns to the body.
c) Urine remains permanently in the kidneys.
d) The body stops producing wastes.



B. Write True or False

Write True (T) or False (F).

1. Excretion is the process by which the body removes harmful and unwanted substances. (T)
2. The kidneys help control the amount of water and salts in the body. (T)
3. The bladder produces urine by filtering the blood. (F)
4. The urinary system includes the kidneys, ureters, bladder, and urethra. (T)
5. Urine contains wastes and extra water removed from the blood. (T)



C. Complete the Following

Fill in the blanks with the correct words.

1. Unwanted substances produced by cells are called **waste** products.
2. The process by which the body removes harmful and unwanted substances is called **excretion**.
3. The urinary system removes waste products from the **blood**.
4. The kidneys help balance the amount of water and **salts** in the body.
5. The kidneys remove wastes from the blood and get rid of them from the body in the form of **urine**.





Healthy Kidneys Happier Lives!

Learn Discover Be Healthy!

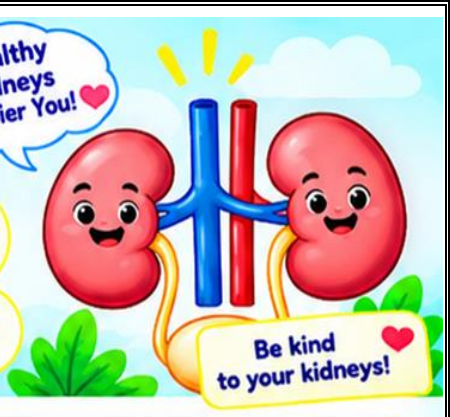
Chapter 4 Circulation and Excretion



Lesson 4-2

How do Kidneys Work?

Healthy kidneys keep your body clean and healthy!



D. What Happens When...?

Think about the result, then answer.

1. Blood containing wastes enters the kidneys?
   The kidneys filter the blood and remove the wastes from it.
2. The kidneys remove wastes and extra water from the blood?
   Urine is formed, and clean blood is returned to the body.
3. Urine leaves the kidneys and enters the ureters?
   It travels through the ureters to the bladder.
4. The bladder fills with urine?
   You feel the need to urinate.
5. The body fails to get rid of urine properly?
   Wastes can build up in the body and may cause health problems.

E. Give Reasons for the Following

Explain why each statement is true.

1. The kidneys are described as filters of the body.
   Because they filter the blood and remove wastes, extra water, and salts from it.
2. The urinary system is important for excretion.
   Because it removes harmful and unwanted substances (wastes) from the body.
3. The kidneys help keep the body healthy.
   Because they remove wastes and extra water, keeping the blood clean and balanced.
4. The bladder is an important part of the urinary system.
   Because it stores the urine until it is ready to leave the body.
5. Wastes produced by body cells must be removed from the body.
   Because they are harmful and can make the body sick if they build up.



Chapter 4 Circulation and Excretion

Lesson 4-3

Various Organs

Different organs work together to keep us alive and healthy!







Try

Answer the following questions.

1 (1) What is the organ called that filters the blood and filters out waste products?

Kidney

(2) What is the liquid waste produced in (1) called?

Urine

(3) How many of (1) are there in the body?

Two



Exercise

Choose the correct answer.

2 (1) Choose 2 organs that are inside the chest of the human body.

A: Stomach
B: Heart
C: Lungs
D: Large intestine
E: Small intestine



3 (2) Among the following organs, which one has one on each side of the body, for a total of two? Choose the correct answer from below.

A: Lungs
B: Stomach
C: Heart
D: Small intestine

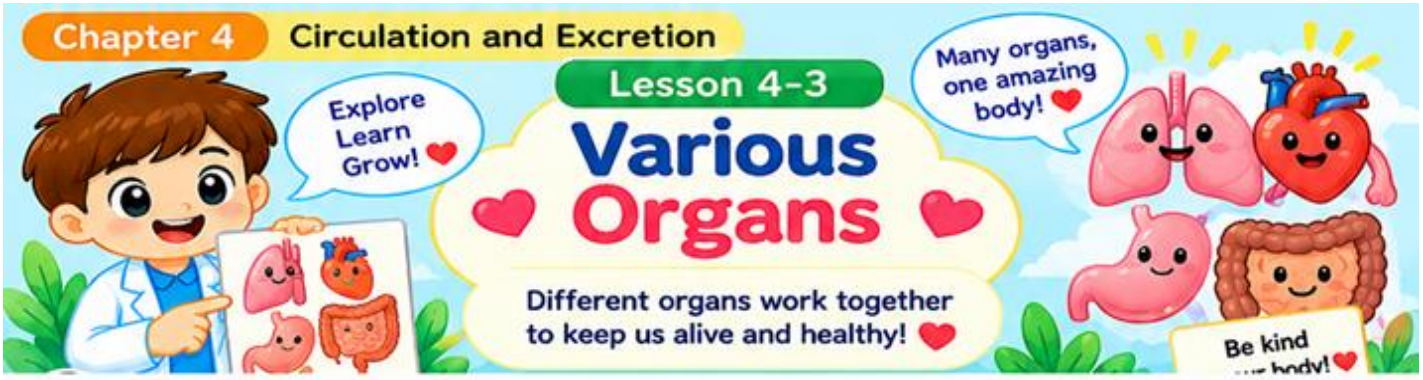


Healthy Bodies Happier Lives!



Learn Discover Be Healthy!





Exercise

Continue the lesson questions.

1 Which statement correctly explains how the kidneys work? Choose the correct answer.

A: It absorbs oxygen from the blood.

B: It pumps blood to the whole body.

C: It takes out waste products from the blood.

Where is the urine produced in the kidneys temporarily stored until it is discharged out of the body?

Urinary bladder

2 Which statement correctly describes where the organs are located? Choose the correct answer.

A: The small intestine is outside the large intestine and surrounds the large intestine.

B: The stomach is connected to the food pipe.

C: The stomach is on the lower side of the abdomen.

D: There are two kidneys on the abdomen side.

Fill in the blank with the correct word.

Each organ is not moving separately, but all are connected by **blood vessels** work together.

